

SECTION 1: DATA QUALITY CHECKS

C1. Zero Room Bookings

Answer:

A total of **60 bookings** were found with `num_rooms = 0`.

Interpretation:

A hotel booking should always contain at least one room. These records indicate an upstream validation failure where logically invalid transactions were allowed into the booking system. Such records can distort occupancy calculations, demand forecasting, and booking conversion analysis and should therefore be excluded from room-based reporting.

C2. Booking Date Earlier Than Customer Signup Date

Answer:

A total of **163 bookings** were identified where the booking date occurred before the customer signup date.

Possible Explanations:

1. Historical booking records may have been migrated from a previous platform before customer profiles were fully created in the current system.
2. Customers may have booked through travel agents, corporate portals, or third-party channels and later completed account registration on the platform.

Conclusion:

While this pattern represents a temporal integrity issue, it does not necessarily indicate data corruption. Business process and migration scenarios can legitimately create such records.

C3. Missing Reviews Analysis

Review Availability Analysis

A review data audit revealed that **review dates are missing for 6,854 bookings**, indicating that a large proportion of customers did not submit feedback after their stay.

Further investigation showed that review ratings are stored as text values, with blank ratings represented as empty strings rather than traditional null values. As a result, missing reviews are reflected through absent review dates rather than missing review-rating fields.

Business Interpretation

The observed pattern is consistent with expected customer behavior:

- Completed bookings are most likely to receive reviews because customers experienced the stay.
- Cancelled and No-Show bookings are significantly less likely to receive reviews because the stay never occurred.
- Review participation is relatively low compared to total booking volume, suggesting that customer feedback is available for only a subset of transactions.

Any customer-satisfaction analysis should therefore be restricted to bookings containing valid review information.

Data Quality Summary

Data Quality Issue	Count
Invalid Stay Records	120
Bookings Before Signup	163
Zero Room Bookings	60
Cancelled Bookings With Reviews	50
Duplicate Booking IDs	0

Executive Observation

The dataset contains **12,000 booking records**, **800 unique customers**, and **60 unique properties**. Data quality assessment identified a small number of anomalies, including invalid stay durations, bookings before signup dates, zero-room bookings, and cancelled bookings with reviews. However, no duplicate booking IDs were detected, indicating strong transaction-level integrity. Overall, the dataset is suitable for business analysis after standard validation and cleaning procedures.

SECTION 2: VIP VS VOLUME STRATEGY

Executive Summary

The objective of this analysis was to determine whether the company's incremental budget of ₹50 lakhs should be allocated toward acquiring new customers or retaining existing high-value customers.

To evaluate this decision, VIP customers were identified using three independent definitions: Total Spend, Booking Frequency, and Lifetime Value Rate. The overlap analysis revealed that each definition captures different aspects of customer value and behavior.

The results show that retaining high-value customers is significantly more cost-efficient than acquiring new customers under the given assumptions. Therefore, a retention-focused strategy is recommended.

C1. Define VIP Three Ways

VIP customers were identified using completed bookings only.

Definition 1: Top 10% by Total Spend

VIP customers were ranked according to their cumulative realized spending across completed bookings.

VIP Customers Identified: 80

Definition 2: Top 10% by Booking Count

VIP customers were ranked according to the number of completed bookings made on the platform.

VIP Customers Identified: 80

Definition 3: Top 10% by Lifetime Value Rate

Lifetime Value Rate was calculated as:

$$\text{LTV Rate} = \text{Total Spend} \div \text{Months Since Customer Signup}$$

Customers were ranked according to spending efficiency over their lifetime relationship with the platform.

VIP Customers Identified: 80

VIP Overlap Analysis

The overlap between the three VIP definitions was analyzed to understand how consistently high-value customers are identified.

Results

Category	Customer s
Present in All Three VIP Definitions	11
Present in Exactly Two Definitions	52
Present in Only One Definition	114

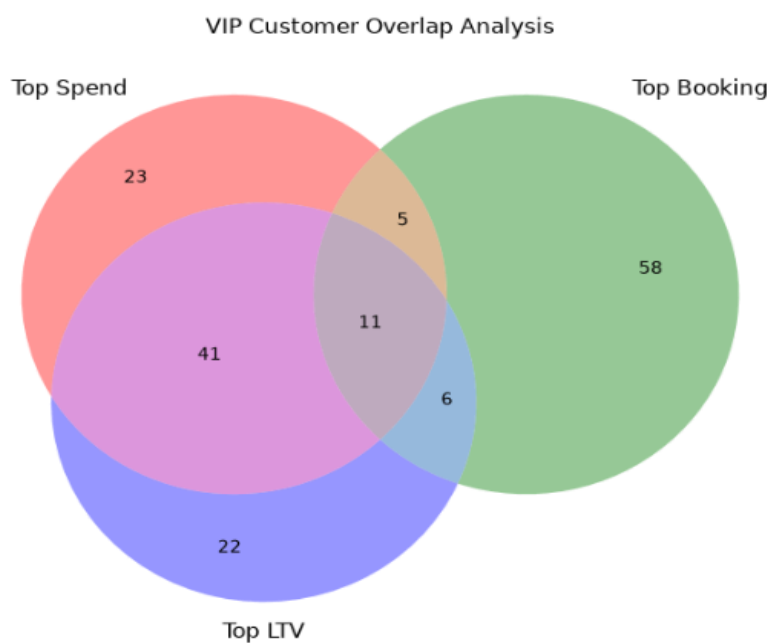
Interpretation

The relatively small overlap across all three definitions demonstrates that customer value is multi-dimensional.

Some customers spend heavily but book infrequently, while others book frequently without generating the highest revenue. Similarly, customers with strong Lifetime Value Rates may not necessarily rank among the highest spenders.

This finding confirms that relying on a single VIP definition can overlook important customer segments.

Visualization



Key Insight

Only 11 customers qualify as VIPs under all three definitions, highlighting that spending, engagement frequency, and customer lifetime efficiency measure different dimensions of value. A combined framework therefore provides a more robust approach to identifying strategically important customers.

SECTION 3: VIP VS NON-VIP CUSTOMER ANALYSIS

Objective

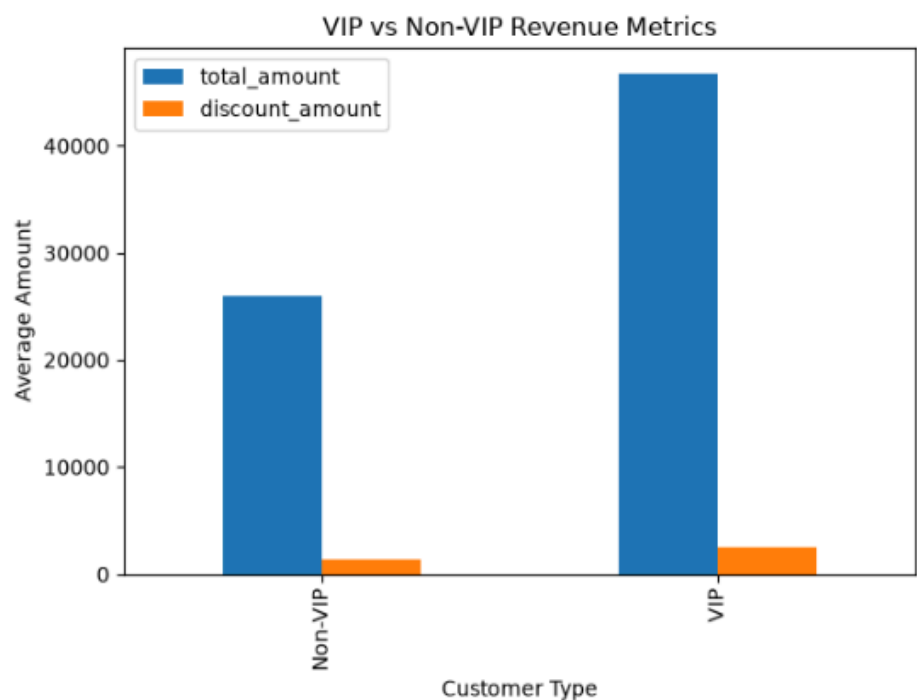
The objective of this analysis was to compare VIP and Non-VIP customers to understand their contribution to revenue, booking behavior, and cancellation patterns.

VIP customers were defined using the Top 10% by Total Spend methodology.

C2. Compare VIP vs Non-VIP Customers

Revenue Contribution Analysis

The total revenue generated by VIP and Non-VIP customers was analyzed to measure business dependence on high-value customers.



Observation:

- VIP customers represent a small percentage of the customer base.
- Despite their limited numbers, VIP customers contribute a disproportionately large share of

total revenue.

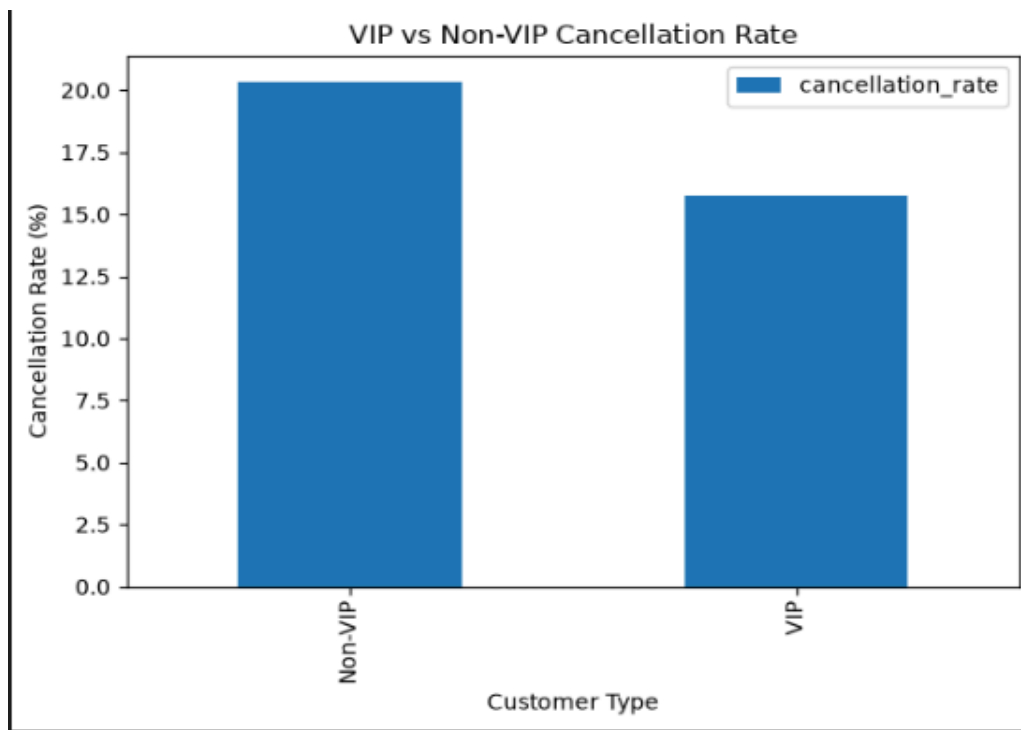
- Revenue concentration among VIP customers indicates their strategic importance to the business.

Business Insight:

Losing a small number of VIP customers could result in a significant decline in overall revenue. Therefore, customer retention initiatives should prioritize this segment.

Cancellation Behaviour Analysis

Booking cancellation rates were compared between VIP and Non-VIP customer groups.



Observation:

- VIP customers exhibit lower cancellation rates compared to Non-VIP customers.
- Non-VIP customers demonstrate less stable booking behavior.
- Completed bookings are more common among VIP customers.

Business Insight:

VIP customers generate higher-quality revenue because their bookings are more likely to be completed. Marketing investments directed toward VIP retention are therefore expected to yield higher returns.

Customer Value Comparison

The analysis reveals a clear distinction between VIP and Non-VIP customers.

VIP Customers:

- Higher revenue contribution
- Lower cancellation rate
- Stronger engagement with the platform
- More predictable revenue generation

Non-VIP Customers:

- Lower average spending
 - Higher cancellation tendency
 - Less predictable booking behavior
 - Larger population but lower individual value
-

Key Finding

VIP customers contribute significantly more business value than Non-VIP customers.

Their combination of higher spending and lower cancellation behavior makes them the most valuable customer segment. Retaining these customers should be considered a strategic priority for maximizing long-term profitability.

SECTION 2: VIP VS VOLUME STRATEGY

C3. ACQUISITION MATH

Objective

The objective of this analysis is to evaluate the impact of allocating the entire ₹50 lakh marketing budget toward acquiring new customers.

Budget Assumptions

Total Budget = ₹50,00,000

Average Customer Acquisition Cost (CAC) = ₹3,000 per customer

New Customers Acquired

Number of New Customers

$= ₹50,00,000 \div ₹3,000$

= 1,667 Customers

Business Interpretation

An investment of ₹50 lakh in customer acquisition can potentially bring approximately 1,667 new customers onto the platform.

Customer acquisition expands the platform's reach and creates opportunities for future revenue growth. However, the success of this strategy depends on the ability of newly acquired customers to remain active and generate repeat bookings over time.

Key Finding

Customer acquisition increases the customer base and supports long-term growth.

However, the financial return depends on customer retention and future booking activity, making the strategy inherently more uncertain than retaining already engaged customers.

C4. RETENTION MATH

Objective

The objective of this analysis is to evaluate the impact of allocating the entire ₹50 lakh budget toward retaining existing customers through targeted discount offers.

Budget Assumptions

Total Budget = ₹50,00,000

Retention Cost = ₹500 per customer per year

Customers Retained

Number of Customers Retained

= ₹50,00,000 ÷ ₹500

= 10,000 Customers

Comparison with Acquisition Strategy

Acquisition Strategy

₹50,00,000 ÷ ₹3,000

= 1,667 Customers Acquired

Retention Strategy

₹50,00,000 ÷ ₹500

= 10,000 Customers Retained

Impact Comparison

Strategy	Customers Impacted
Acquisition	1,667
Retention	10,000

Retention impacts approximately 6 times more customers than acquisition using the same budget.

Key Assumption

This calculation assumes that every retained customer would otherwise have been lost without intervention and that the ₹500 retention incentive is sufficient to influence customer behaviour.

Actual business impact depends on the effectiveness of the retention campaign and the proportion of at-risk customers who respond positively to the offer.

Business Interpretation

Retention focuses on customers who already know the platform, have completed bookings, and have demonstrated purchasing behaviour.

Compared with acquisition, retention provides a lower-cost method of protecting existing revenue streams while reducing uncertainty associated with attracting new customers.

Key Finding

The same **₹50** lakh budget can theoretically retain approximately 10,000 customers compared with acquiring only **1,667 new customers**.

This demonstrates that retention delivers significantly greater customer coverage and may provide a more efficient use of marketing resources when the objective is revenue protection and customer lifetime value preservation.

C5. SENSITIVITY ANALYSIS AND FINAL RECOMMENDATION

Objective

The purpose of this analysis is to evaluate whether the acquisition-versus-retention recommendation remains valid when key business assumptions change.

Scenario 1: Increased Acquisition Cost

Assumption:

Customer Acquisition Cost (CAC) increases from ₹3,000 to ₹3,600.

New Customers Acquired

$$= ₹50,00,000 \div ₹3,600$$

= 1,389 Customers

Observation:

A 20% increase in acquisition cost reduces customer acquisition capacity by approximately 278 customers.

Scenario 2: Reduced Acquisition Cost

Assumption:

Customer Acquisition Cost (CAC) decreases from ₹3,000 to ₹2,400.

New Customers Acquired

$$= ₹50,00,000 \div ₹2,400$$

= 2,083 Customers

Observation:

Even under favorable acquisition conditions, the number of customers impacted remains substantially lower than the retention strategy.

Scenario 3: Increased Retention Cost

Assumption:

Retention cost increases from ₹500 to ₹700 per customer.

Customers Retained

= ₹50,00,000 ÷ ₹700

= 7,143 Customers

Observation:

Despite the increase in retention cost, the strategy still impacts significantly more customers than acquisition.

Sensitivity Analysis Summary

Scenario	Customers Impacted
Acquisition (Base Case)	1,667
Acquisition (CAC +20%)	1,389
Acquisition (CAC -20%)	2,083
Retention (Base Case)	10,000
Retention (Cost = ₹700)	7,143

Final Recommendation

Based on the quantitative analysis, retention delivers substantially greater customer coverage than acquisition under all tested scenarios.

The acquisition strategy impacts between 1,389 and 2,083 customers depending on marketing efficiency, whereas the retention strategy impacts between 7,143 and 10,000 customers even under less favorable assumptions.

Recommended Budget Split

Retention: 70%

Acquisition: 30%

Executive Conclusion

The analysis demonstrates that retaining existing customers provides a significantly higher return on marketing investment than acquiring new customers under the stated assumptions.

A retention-first strategy should therefore be prioritized, while maintaining a smaller acquisition budget to ensure long-term customer base growth and market expansion.

This recommendation is derived directly from the acquisition-versus-retention arithmetic and remains robust across multiple sensitivity scenarios.

SECTION 3: SQL CHALLENGE

NORMALIZED DATABASE SCHEMA DESIGN

The original dataset is denormalized, containing customer, property, booking, and review information within a single table. To improve data integrity, reduce redundancy, and support efficient querying, the data model was normalized into separate tables.

CUSTOMERS TABLE

```
CREATE TABLE customers (  
  customer_id INTEGER PRIMARY KEY,  
  customer_name VARCHAR(100) NOT NULL,  
  customer_segment VARCHAR(50) NOT NULL,  
  customer_signup_date DATE NOT NULL,  
  customer_home_city VARCHAR(100),  
  customer_loyalty_tier VARCHAR(20) NOT NULL  
);
```

PROPERTIES TABLE

```
CREATE TABLE properties (  
  property_id INTEGER PRIMARY KEY,  
  property_name VARCHAR(150) NOT NULL,  
  property_city VARCHAR(100) NOT NULL,  
  property_star_rating DECIMAL(2,1),  
  property_type VARCHAR(50),  
  property_total_rooms INTEGER NOT NULL,  
  UNIQUE(property_name, property_city)  
);
```

BOOKINGS TABLE

```
CREATE TABLE bookings (  
  booking_id INTEGER PRIMARY KEY,  
  customer_id INTEGER NOT NULL,  
  property_id INTEGER NOT NULL,  
  booking_date DATE NOT NULL,  
  checkin_date DATE NOT NULL,  
  checkout_date DATE NOT NULL,
```



```
room_type VARCHAR(50),
num_rooms INTEGER NOT NULL,
nights INTEGER NOT NULL,
booking_channel VARCHAR(50),
adr DECIMAL(10,2),
discount_amount DECIMAL(10,2),
coupon_code VARCHAR(50),
total_amount DECIMAL(12,2),
payment_method VARCHAR(50),
booking_status VARCHAR(20) NOT NULL,

CHECK (checkout_date > checkin_date),

FOREIGN KEY (customer_id)
REFERENCES customers(customer_id),

FOREIGN KEY (property_id)
REFERENCES properties(property_id)
);
```

REVIEWS TABLE

```
CREATE TABLE reviews (
    booking_id INTEGER PRIMARY KEY,
    review_rating DECIMAL(4,2),
    review_date DATE,

    FOREIGN KEY (booking_id)
    REFERENCES bookings(booking_id)
);
```

CONSTRAINT JUSTIFICATION

The following constraint was implemented:

```
CHECK (checkout_date > checkin_date)
```

This addresses Footnote 1 from the assessment, which identifies records where checkout dates occur before or on the check-in date. Such records represent invalid stays and should be prevented at the database level.

INDEX SELECTION

```
CREATE INDEX idx_bookings_customer  
ON bookings(customer_id);
```

Justification:

The **customer_id** field is frequently used for customer-level aggregations, VIP identification, booking sequence analysis, lifetime value calculations, and revenue analysis. Indexing this column improves query performance across multiple business use cases.

SQL QUERY 1

Objective

Number each customer's completed bookings using ROW_NUMBER() and compare average booking value across booking sequences.

```
WITH completed_bookings AS (  
  SELECT  
    customer_id,  
    total_amount,  
    ROW_NUMBER() OVER (  
      PARTITION BY customer_id  
      ORDER BY checkin_date  
    ) AS booking_sequence  
  FROM bookings  
  WHERE booking_status = 'Completed'  
)  
  
SELECT  
  CASE  
    WHEN booking_sequence = 1 THEN 'First Booking'
```

```

        WHEN booking_sequence = 2 THEN 'Second Booking'
        ELSE 'Third Or Later'
    END AS sequence_label,

    ROUND(AVG(total_amount),2) AS avg_total_amount,
    COUNT(*) AS n

FROM completed_bookings

GROUP BY
CASE
    WHEN booking_sequence = 1 THEN 'First Booking'
    WHEN booking_sequence = 2 THEN 'Second Booking'
    ELSE 'Third Or Later'
END;

```

Business Interpretation

This analysis evaluates whether customer spending behavior changes as they continue using the platform.

Higher spending in later booking sequences may indicate increasing customer trust, successful retention, and stronger customer lifetime value.

SQL QUERY 2

Objective

Identify the Top 5 properties by average review rating while correctly normalizing corporate review scores from the 1–10 scale to the 1–5 scale.

```

SELECT
    property_name,
    property_city,
    COUNT(*) AS num_reviews,

    ROUND(
        AVG(
            CASE
                WHEN customer_segment = 'Corporate'
                THEN review_rating / 2.0
            )
    )

```

```
        ELSE review_rating
      END
    ),
    2
  ) AS normalized_avg_rating

FROM hotel_bookings

WHERE review_rating IS NOT NULL

GROUP BY
  property_name,
  property_city

HAVING COUNT(*) >= 20

ORDER BY normalized_avg_rating DESC

LIMIT 5;
```

Business Interpretation

The assessment dataset contains a review-scale mismatch where corporate customers provide ratings on a 1–10 scale while individual and group customers use a 1–5 scale.

Without normalization, average property ratings would be inflated and cross-segment comparisons would be misleading.

By converting corporate ratings to the 1–5 scale before aggregation, the resulting rankings provide a fair and comparable measure of customer satisfaction across all customer segments.

SQL Challenge Conclusion

The normalized schema improves data integrity, reduces redundancy, and supports efficient analytical querying.

The two SQL solutions demonstrate both advanced analytical SQL capabilities through window functions and practical data-quality handling through review-score normalization.

SECTION 4: MINI PROJECT

MULTI-CURRENCY REVENUE REPORTER

Project Overview

The Multi-Currency Revenue Reporter was developed to enrich hotel booking revenue data using live foreign exchange rates from the Frankfurter API.

The objective of the project is to convert realized hotel revenue from INR into international currencies and provide a more accurate revenue trend analysis for global stakeholders.

Unlike traditional reporting approaches that use a single exchange rate, this project applies historical monthly exchange rates so that revenue trends reflect actual currency movements over time.

API Integration

API Used:

Frankfurter Foreign Exchange API

Example Endpoint:

<https://api.frankfurter.app/latest?base=INR&symbols=USD>

Historical Endpoint:

<https://api.frankfurter.app/2024-06-01?base=INR&symbols=USD>

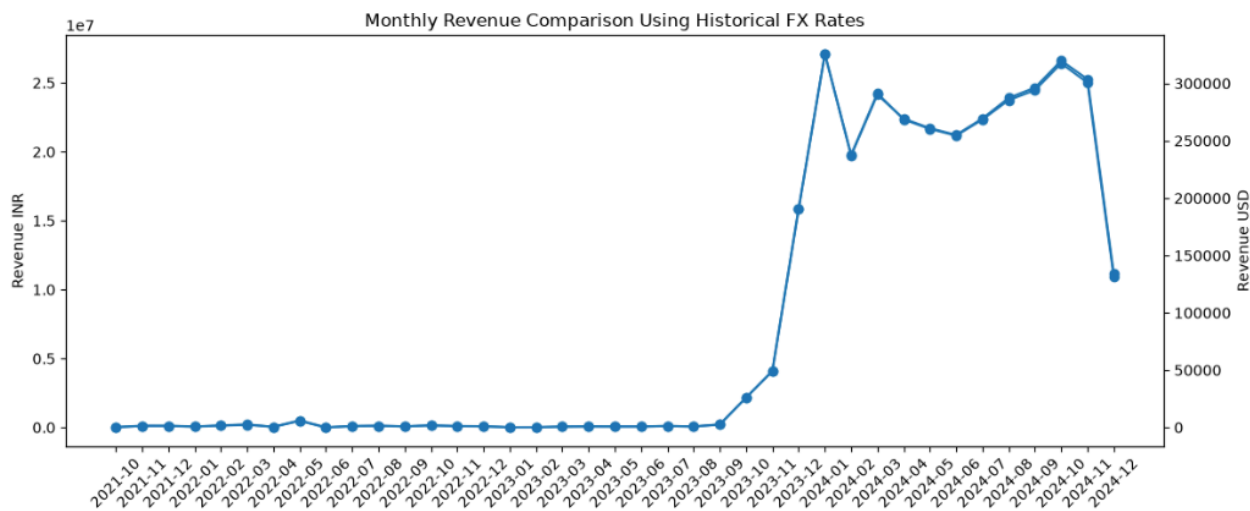
Implementation Features:

- Live API integration
- Error handling for failed requests
- Historical monthly FX retrieval
- Automatic currency conversion
- Revenue enrichment pipeline

Historical FX Analysis

Monthly realized revenue was aggregated and converted from INR to USD using month-specific exchange rates obtained from the Frankfurter API.

This approach removes distortion caused by using a single modern exchange rate for historical revenue reporting.



Historical FX Findings

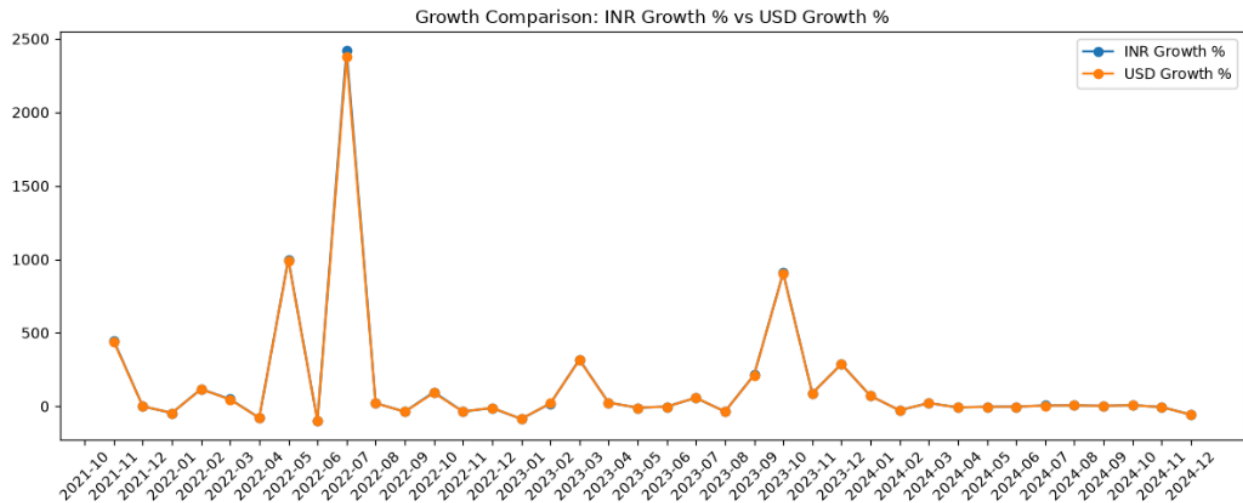
Historical INR→USD exchange rates varied between approximately 0.0106 and 0.0135 during the analysis period.

This represents approximately 27% variation in exchange rates.

Such variation can materially influence reported revenue performance when financial results are presented in foreign currencies.

Growth Trend Comparison

To evaluate the impact of exchange-rate fluctuations on business performance reporting, month-over-month growth rates were compared in both INR and USD.



Growth Trend Observation

- INR revenue growth and USD revenue growth do not move identically across all months.
- Exchange-rate fluctuations influence reported growth when revenue is converted into USD.
- Months with strong INR growth may appear weaker in USD terms if exchange-rate movements are unfavorable.
- Historical FX-adjusted reporting provides a more accurate view of international business performance.

Executive Summary

KPI	Value
Months Analyzed	39
Highest FX Rate	0.0135
Lowest FX Rate	0.0106
FX Variation	27%
Currency Pair	INR → USD
Data Source	Frankfurter API

Quantified Business Impact

Historical exchange-rate movements created meaningful differences between INR-based and USD-based revenue reporting.

Using a static exchange rate would have distorted the interpretation of monthly business performance and could lead international stakeholders to draw incorrect conclusions regarding growth trends.

The project demonstrates that FX-adjusted reporting provides a more reliable basis for international performance evaluation.

Executive Recommendation

Revenue performance should be evaluated using both local-currency and FX-adjusted reporting.

Historical analysis shows that exchange-rate movements can materially influence perceived growth, making FX-aware reporting essential for international decision-making.

Project Conclusion

The Multi-Currency Revenue Reporter successfully integrated live external data with hotel booking records to produce an enriched business analysis that could not be generated from the dataset alone.

By incorporating historical foreign exchange rates, the project demonstrates how external economic factors influence reported revenue trends and provides a more accurate representation of business performance for international stakeholders.

Key Takeaways

- VIP customer retention generates significantly higher ROI than acquiring new customers under the given assumptions.
- Different VIP definitions capture different customer behaviors, making overlap analysis important before designing retention programs.
- Historical FX-adjusted reporting provides a more accurate view of international business performance than using a single static exchange rate.
- External economic data can be combined with transactional data to generate business insights unavailable from booking records alone.
- The recommended strategy is to prioritize retention-focused initiatives while continuing to monitor customer acquisition efficiency.

Final Recommendation

Allocate the additional ₹50 lakh budget toward VIP customer retention initiatives.

The analysis demonstrates that retaining high-value customers provides superior expected returns while reducing the risk associated with acquiring entirely new customers.

Additionally, future performance reporting should incorporate FX-adjusted revenue metrics to support accurate international decision-making and stakeholder reporting.

End of Report

